

MARCO ANTONIO GONZALEZ JUNIOR

m@rco.sh · +55 (11) 91577-7715 · São Paulo, Brazil · linkedin.com/in/marcoantoniogonzalezjunior

Solutions Architect | Cloud Engineer | Artificial Intelligence Specialist

PROFESSIONAL SUMMARY

Solutions Architect with 16+ years building scalable enterprise systems and 10+ years in AI/ML. Led technical teams delivering solutions that generated 30% revenue increases and 200% improvement in customer retention across financial, retail, and digital marketing. Currently specializing in advanced generative AI: custom model development, fine-tuning, RAG, agentic workflows, multi-agent systems, MCP-based integrations, and enterprise AI governance.

SKILLS

Software Architecture: Domain-Driven Design, Clean Architecture, Event-Driven Architecture, Microservices, Enterprise Integration Patterns

Solutions Architecture: Cloud-Native Design, API Strategy, System Integration, Scalability Patterns, Cost Optimization, Vendor Evaluation, Technology Roadmaps

Cloud & DevOps: AWS/Azure/GCP, Databricks, Kubernetes, Docker, Terraform, CI/CD, Infrastructure as Code, Serverless

AI & Machine Learning: Generative AI, LLMs, Fine-tuning, RAG Systems, Knowledge Graphs, Multi-Agent Systems (LangGraph, Google ADK, CrewAI), Model Context Protocol (MCP), MLOps, LLMOps

Technical Leadership: Hands-on Leadership, Technology Strategy, Innovation, Evangelism, Digital Transformation

EXPERIENCE

Sabesp

Lead AI Specialist

São Paulo

September 2025 – Present

Leading AI strategy and engineering at the world's third-largest water utility, driving Generative AI enablement, agentic systems, and digital transformation across operational and customer-facing domains.

- Leading agentic AI development on the Databricks platform, building production-grade agents on Mosaic AI Agent Framework and integrating Databricks Genie spaces for natural-language access to Unity Catalog gold tables across 50+ business KPIs.
- Architected and developed MCP (Model Context Protocol) servers and clients to expose Databricks Genie, enterprise data sources, and internal tools as standardized capabilities, enabling reusable agent integrations across Microsoft Teams bots, Claude, and internal AI clients.
- Designed and led the rebuild of the Customer Service Quality Audit platform (formerly Speech Analytics): a LangGraph multi-agent system that transcribes outsourced call-center recordings via Whisper and audits them against Sabesp's service standards, directly impacting NPS and contractor governance.
- Delivered a conversational KPI agent for C-level executives on Microsoft Teams, integrating Azure Bot Service, Adaptive Cards, Azure Entra ID (OAuth M2M), and Databricks Genie via MCP – providing real-time, natural-language access to financial, operational, and commercial indicators.
- Building agents across multiple frameworks (LangGraph, Google ADK, Mosaic AI Agent Framework) and evaluating trade-offs between orchestration patterns, tool-calling models, and agent runtimes for enterprise-scale workloads.
- Architecting a centralized AI Gateway providing model access with fallbacks, spend tracking, rate limiting, security policies, and prompt caching across Azure OpenAI, AWS Bedrock, and Databricks-hosted foundation models.
- Implementing MLOps/LLMOps CI/CD pipelines on Databricks and Azure, serving foundation and fine-tuned models that power specialized agentic systems for customer support, code review, and operational analytics.
- Partnering with internal stakeholders to translate operational challenges in water and sanitation infrastructure into AI roadmaps, establishing adoption patterns, governance, and clear paths to production for generative AI solutions.

Wipro

Senior Solutions Architect – Artificial Intelligence

São Paulo

September 2024 – September 2025

Led AI-driven legacy modernization initiatives using knowledge graphs and LLMs to transform enterprise COBOL systems into modern technology stacks.

- Designed intelligent legacy modernization platform using Neo4j knowledge graphs, LangGraph, and AWS Bedrock to automate COBOL application transformation, delivering 85%+ reduction in migration timelines.
- Engineered production RAG systems combining semantic search (Pinecone, Weaviate) and knowledge graphs, achieving 90% fidelity in preserving business logic during COBOL to Java/.NET migration.
- Led team of 8 engineers developing intelligent AI workflows, enabling automated extraction of functional requirements from enterprise legacy systems – reducing manual documentation time by 90% across millions of lines of code.

IBM

Senior Solutions Architect – Artificial Intelligence

São Paulo

July 2022 – September 2024

Architected advanced AI solutions for financial services clients, including foundation model fine-tuning and hybrid RAG systems with continuous learning capabilities.

- Fine-tuned custom LLMs specialized to business domains through fine-tuning pipelines and RAG knowledge bases using PyTorch and TensorFlow, achieving 75% automated resolution rate and 25% improvement in customer satisfaction.

- Designed multi-cloud MLOps platform (AWS, Azure) with automated continuous learning pipelines, ensuring 99% uptime while serving 10M+ monthly interactions; standardized deployments through Docker, Kubernetes and Terraform, cutting deployment times by 80%.
- Developed comprehensive model evaluation framework integrating BERTScore semantic analysis and human-in-the-loop feedback, enabling continuous model refinement under strict security and compliance constraints.

banQi

Senior Solutions Architect

São Paulo
June 2019 – July 2022

Led technical architecture and engineering teams at fast-growing fintech startup, promoted from Technical Lead to Principal Engineer, driving system scalability and strategic technical direction.

- Architected microservices platform using Docker, Kubernetes, TypeScript and Go supporting 10x traffic growth, reducing system response times by 65% and achieving 99.95% uptime.
- Built ML-powered fraud detection system with real-time scoring, reducing fraudulent transactions by 40% and false positives by 25%.
- Unified system architecture implementing Domain-Driven Design and serverless computing (AWS Lambda, Node.js, TypeScript); led containerization across 15+ applications using Helm, ArgoCD, and GitOps workflows, reducing deployment times by 80%.
- Built and managed cross-functional engineering team of 12+ developers, implementing agile practices with GitHub Actions CI/CD and increasing delivery velocity by 45%.

Petlove

Tech Lead

São Paulo
May 2015 – June 2019

Led technical architecture for Brazil's largest pet supplies e-commerce platform, driving digital transformation initiatives.

- Architected prime membership platform using Ruby on Rails, Elixir, and PostgreSQL, achieving 30% of monthly revenue (\$2M+) and 500% increase in customer repurchase rate.
- Led ERP system modernization using Python and distributed warehouse management, enabling multi-state expansion and supporting 200% business growth.
- Built Business Intelligence team of 6 engineers; implemented big data and ML solutions with ETL pipelines that improved inventory forecasting accuracy by 85% and reduced procurement costs by 15%.

Jüssi

Solutions Architect

São Paulo
November 2012 – May 2015

Architected digital marketing automation solutions as Facebook Marketing Partner and Google Partner for enterprise clients.

- Led Facebook Marketing Partner certification, collaborating with Facebook Engineering using GraphQL APIs to deliver marketing automation for biggest e-commerces in Brazil.
- Architected Google Ads automation platforms using Java and RESTful APIs, managing \$5M+ monthly ad spend and achieving 40% improvement in ROAS; built real-time bidding optimization that reduced customer acquisition costs by 25% across 5k+ enterprise campaigns.

Editora Abril

Senior Software Engineer

São Paulo
September 2008 – November 2012

- Developed large-scale content management systems for Brazil's leading media and entertainment publisher.
- Built Veja SP platform using HTML5, CSS3, JavaScript, and MySQL, serving 500K+ monthly users with 99.9% uptime.
 - Architected enterprise content management system using Java and SOA, supporting 20+ publications and 200+ concurrent editors.

PAST EXPERIENCES

Serasa Experian (Java Enterprise Architect), Western Union (Senior Software Engineer), Toyota do Brasil (Software Engineer), Wdotnet Internet (Owner), Flyever Electronics (Software Engineer)

EDUCATION

Faculdade de Informática e Administração Paulista

Faculdade Impacta de Tecnologia

Universidade Federal de São Carlos

MBA Data Science (2015–2016)
BSc Information Systems (2011–2015)
BSc Materials Engineering (1996–2000)

CERTIFICATIONS

AWS: Solutions Architect Professional (2025), Data Engineer Associate (2025), ML Engineer (2024), ML Specialty (2024), AI Practitioner (2024), Solutions Architect Associate (2022), Developer (2022)
Specializations: Stanford Machine Learning and Deep Learning (2024), IBM AI Engineering Professional (2024), Data Science (2015), Recommender Systems (2014)

LANGUAGES

Portuguese: Native

English: Fluent

Spanish: Conversational